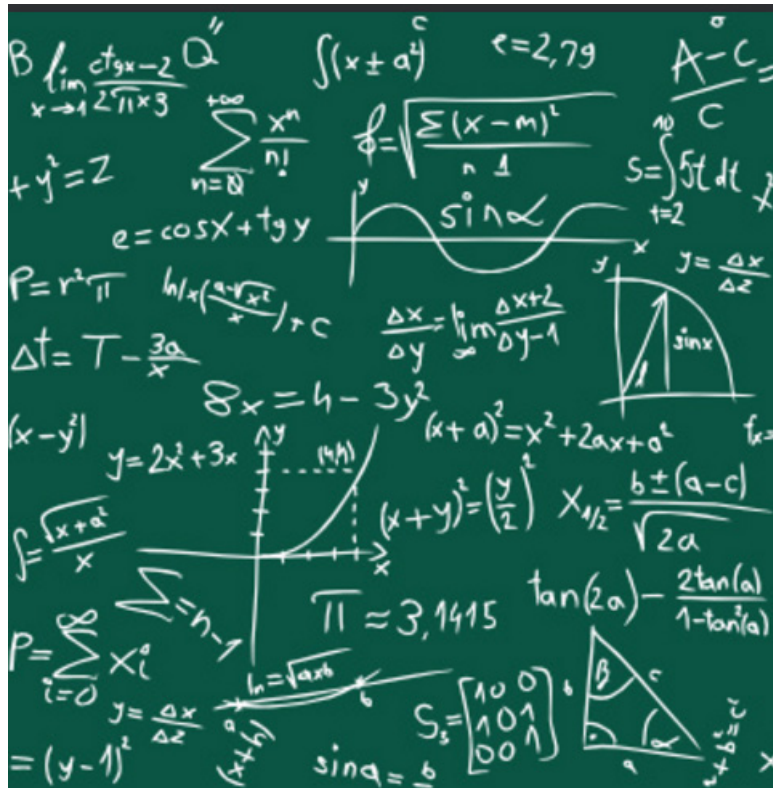


## Exercises: complex numbers



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** “Esercitazioni Analisi matematica 1”, Marco Bramanti, Esculapio.

### Exercise 1

Write the following complex numbers  $z$  in algebraic form, identifying their real part and imaginary part. Then, for each of them, find the modulus, the conjugate and the reciprocal.

$$z = \frac{2+5i}{1+i} \qquad z = \frac{2+i}{1-3i}$$

$$z = \frac{2+5i}{1+i} \cdot \frac{1-i}{1-i} = \frac{7+3i}{2} = \frac{7}{2} + i\frac{3}{2}$$

from which

$$\Re z = \frac{7}{2}, \Im z = \frac{3}{2}, |z| = \sqrt{\frac{49}{4} + \frac{9}{4}} = \frac{\sqrt{58}}{2}, \bar{z} = \frac{7}{2} - i\frac{3}{2}.$$

Finally

$$z^{-1} = \frac{1}{z} \cdot \frac{\bar{z}}{\bar{z}} = \frac{\bar{z}}{|z|^2} = \frac{7-3i}{2} \cdot \frac{4}{58} = \frac{7}{29} - i\frac{3}{29}.$$

$$z = \frac{2+i}{1-3i} \cdot \frac{1+3i}{1+3i} = \frac{-1+7i}{10} = -\frac{1}{10} + i\frac{7}{10}$$

from which

$$\Re z = -\frac{1}{10}, \Im z = \frac{7}{10}, |z| = \sqrt{\frac{1}{100} + \frac{49}{100}} = \frac{\sqrt{2}}{2}, \bar{z} = -\frac{1}{10} - i\frac{7}{10}.$$

Finally

$$z^{-1} = \frac{1}{z} \cdot \frac{\bar{z}}{\bar{z}} = \frac{\bar{z}}{|z|^2} = -\frac{1+7i}{10} \cdot 2 = -\frac{1}{5} - i\frac{7}{5}.$$

### Exercise 2

Determine all the complex numbers  $z$  that satisfy:

$$|z| \leq |z - 2i|$$

It is convenient to use the form  $z = x + iy$ ,  $x, y \in \mathbb{R}$ .

$$|x + iy| \leq |x + i(y - 2)|$$

$$|x + iy|^2 \leq |x + i(y - 2)|^2$$

$$x^2 + y^2 \leq x^2 + (y - 2)^2$$

$$0 \leq -4y + 4$$

$$y \leq 1.$$

Representing the solutions in the Cartesian plane, we obtain a closed half-plane.

### Exercise 3

Determine all the complex numbers  $z$  that satisfy:

$$|z - (3 + i)| \leq 2$$

This inequality is satisfied by those complex numbers  $z = x + iy$  which, in the Gauss plane (complex plane), are represented by the points  $P(x, y)$  whose distance from the point  $(3, 1)$  is less than or equal to 2.

That is, all and only the points of the **closed disk** with center  $C(3, 1)$  and radius 2 (careful: it is not the circle, but the disk, i.e., the interior together with the boundary). The circle has equation:

$$\gamma: (x-3)^2 + (y-1)^2 = 4$$

while the closed disk has equation:

$$(x-3)^2 + (y-1)^2 \leq 4$$

Alternatively, the inequality can be handled through the substitution  $z = x + iy$ , obtaining

$$|x + iy - 3 - i| \leq 2$$

in which, factoring out the imaginary unit,

$$|x - 3 + i(y - 1)| \leq 2$$

Applying the definition of the modulus of a complex number, we have

$$\sqrt{(x-3)^2 + (y-1)^2} \leq 2$$

Since we can square both non-negative sides, we finally obtain:

$$(x-3)^2 + (y-1)^2 \leq 4$$

#### Exercise 4

Determine all the complex numbers  $z$  that satisfy:

$$(1-i)z - (1+i)\bar{z} = i$$

$$(1-i)(x+iy) - (1+i)(x-iy) = i$$

$$x + y - ix + iy - x - y - ix + iy = i$$

$$i(2y - 2x) = i$$

$$2y - 2x = 1.$$

Representing the solutions in the Cartesian plane, we obtain a line.

### Exercise 5

Determine all the complex numbers  $z$  that satisfy:

$$|z + \bar{z}| + |z - \bar{z}| \leq 2$$

$$|z + \bar{z}| + |z - \bar{z}| \leq 2$$

$$2|\Re z| + 2|\Im z| \leq 2$$

$$|x| + |y| \leq 1.$$

Representing the solutions in the Cartesian plane, we obtain a closed square with vertices  $\pm 1, \pm i$ .

### Exercise 6

Determine all the complex numbers  $z$  that satisfy:

$$\Im(z) - |z + \bar{z}|^2 < 1$$

$$\Im(z) - |z + \bar{z}|^2 < 1$$

$$\Im(z) - 4|\Re z|^2 < 1$$

$$y < 4x^2 + 1.$$

Representing the solutions in the Cartesian plane, we obtain the open region below the parabola  $y = 4x^2 + 1$ .

### Exercise 7

Determine all the complex numbers  $z$  that satisfy:

$$\begin{cases} z\bar{z} \leq 2 \\ z + \bar{z} \leq 2 \end{cases}$$

$$\begin{cases} |z|^2 \leq 2 \\ 2\Re z \leq 2 \end{cases} \quad \begin{cases} x^2 + y^2 \leq 2 \\ x \leq 1 \end{cases}$$

Representing the solutions in the Cartesian plane, we obtain the part of the plane common to the closed disk with center at the origin and radius  $\sqrt{2}$  and to the closed half-plane to the left of the vertical line  $x = 1$ .

### Exercise 8

The system in  $\mathbb{C}$

$$\begin{cases} z^2 + \bar{z}^2 = 0 \\ |\Re(z)| + |\Im(z)| = 1 \end{cases}$$

has:

- (a) two solutions    (b) three solutions  
(c) four solutions    (d) none of the other answers is correct

Setting  $z = x + iy$ ,  $x, y \in \mathbb{R}$ , we have

$$\begin{cases} (x + iy)^2 + (x - iy)^2 = 0 \\ |x| + |y| = 1 \end{cases}$$

$$\begin{cases} x^2 - y^2 + 2ixy + x^2 - y^2 - 2ixy = 0 \\ |x| + |y| = 1 \end{cases}$$

$$\begin{cases} x^2 = y^2 \\ |x| + |y| = 1 \end{cases} \quad \begin{cases} |x| = |y| \\ |x| + |y| = 1 \end{cases} \quad \begin{cases} |x| = 1/2 \\ |y| = 1/2 \end{cases}$$

We obtain the four points  $(1 + i)/2$ ,  $(-1 + i)/2$ ,  $(1 - i)/2$ ,  $(-1 - i)/2$ . The correct answer is (c).

### Exercise 9

The solutions  $z = x + iy$  of the complex equation

$$z^2 - z\bar{z} + iz = -3 + 2i$$

satisfy

$$(a) \left(x + \frac{1}{6}\right)^2 + \left(y + \frac{1}{4}\right)^2 = 1 \quad (b) y = \frac{2}{3}x + 1$$

$$(c) \left(x - \frac{1}{6}\right)^2 + \left(y - \frac{1}{4}\right)^2 = 1 \quad (d) y = \frac{3}{2}x$$

$$(x + iy)^2 - (x^2 + y^2) + ix - y = -3 + 2i$$

$$-2y^2 - y + i(x + 2xy) = -3 + 2i$$

$$\begin{cases} 2y^2 + y - 3 = 0 \\ x + 2xy = 2 \end{cases} \quad \begin{cases} y = 1 \vee y = -3/2 \\ x = 2/(1 + 2y) \end{cases}$$

We obtain the two points  $(2/3) + i$ ,  $-1 - i(3/2)$ , which lie on the line  $y = (3/2)x$ .  
The correct answer is (d).

### Exercise 10

Find the modulus and the principal argument of

$$(a) z = -1 - i\sqrt{3}, \quad (b) z = -4i.$$

(a) We have  $|z| = \sqrt{1+3} = 2$ . The principal argument  $\text{Arg } z$  is the unique  $\vartheta \in [-\pi, \pi)$  such that

$$\cos \vartheta = \Re z / |z| = -1/2, \quad \sin \vartheta = \Im z / |z| = -\sqrt{3}/2$$

hence  $\text{Arg } z = -2\pi/3$ .

(b) We have  $|z| = 4$ . The principal argument  $\text{Arg } z$  is the unique  $\vartheta \in [-\pi, \pi)$  such that

$$\cos \vartheta = \Re z / |z| = 0, \quad \sin \vartheta = \Im z / |z| = -1$$

hence  $\text{Arg } z = -\pi/2$ .

### Exercise 11

Let

$$w = \frac{z+1-i}{z+i}$$

Determine, locating them in the Gauss plane, the complex numbers  $z$  for which  $w$  is a purely imaginary number.

The complex number  $w$  is defined for  $z \neq -i$ . Moreover,

$$w = \frac{z+1-i}{z+i} \cdot \frac{z-i}{z-i} = \frac{z^2 + z - 1 + i(-2z - 1)}{z^2 + 1}$$

That is, separating the real part of  $w$  from its imaginary part:

$$w = \frac{z^2 + z - 1}{z^2 + 1} + i \frac{-2z - 1}{z^2 + 1}$$

The complex number  $w$  is purely imaginary when  $\Re(w) = 0$ , i.e.,

$$\Re(w) = \frac{z^2 + z - 1}{z^2 + 1} = 0$$

This rational equation, under the condition  $z \neq \pm\sqrt{-1}$ , i.e.,  $z \neq \pm i$ , gives the real numbers

$$z_1 = -\frac{1}{2} + \frac{\sqrt{5}}{2}, \quad z_2 = -\frac{1}{2} - \frac{\sqrt{5}}{2}$$

which, in the Gauss plane, can be represented as two points on the real axis. In conclusion,  $w$  is purely imaginary for the two values of  $z$  found.

### Exercise 12

Let  $z = \sqrt{3} - 3i$ . The number of solutions  $w$  with  $\Im(w) < 0$  of the equation

$$w^6 = |z|(z + \bar{z})$$

is:

- (a) two    (b) three  
(c) zero    (d) none of the other answers is correct

We have  $|z| = \sqrt{12}$ ,  $z + \bar{z} = 2\sqrt{3}$ , hence

$$w^6 = 12.$$

The complex sixth roots of a positive real number are six, pairwise opposite, and two of them are real. Of the remaining four, two have positive imaginary part and two have negative imaginary part. The correct answer is (a). We nevertheless want to determine their values. Setting  $w = r(\cos \vartheta + i \sin \vartheta)$  and taking into account that  $|12| = 12$ ,  $\arg 12 = 2k\pi$ ,  $k \in \mathbb{Z}$ , we obtain

$$\begin{cases} r^6 = 12 \\ 6\vartheta = 2k\pi \end{cases} \quad \begin{cases} r = 12^{1/6} \\ \vartheta = k\pi/3 \end{cases}$$

We therefore have the six roots

$$w_k = 12^{1/6}(\cos k\pi/3 + i \sin k\pi/3), \quad k = 0, 1, 2, 3, 4, 5,$$

i.e.,

$$w_0 = 12^{1/6}, \quad w_1 = 12^{1/6}(1/2 + i\sqrt{3}/2), \quad w_2 = 12^{1/6}(-1/2 + i\sqrt{3}/2)$$

$$w_3 = -12^{1/6}, \quad w_4 = 12^{1/6}(-1/2 - i\sqrt{3}/2), \quad w_5 = 12^{1/6}(1/2 - i\sqrt{3}/2).$$

The two with negative imaginary part are  $w_4$  and  $w_5$ .

### Exercise 13

Solve the following system in the unknown  $z \in \mathbb{C}$ .

$$\begin{cases} z^6 = i \\ \Re(z)\Im(z) < 0 \end{cases}$$

We first solve the equation  $z^6 = i$ . Setting  $z = r(\cos \vartheta + i \sin \vartheta)$  and taking into account that  $|i| = 1$ ,  $\arg(i) = \pi/2 + 2k\pi$ ,  $k \in \mathbb{Z}$ , we obtain

$$\begin{cases} r^6 = 1 \\ 6\vartheta = \pi/2 + 2k\pi \end{cases} \quad \begin{cases} r = 1 \\ \vartheta = \pi/12 + k\pi/3. \end{cases}$$

We therefore have the six roots

$$z_k = \cos((\pi + 4k\pi)/12) + i \sin((\pi + 4k\pi)/12), \quad k = 0, 1, 2, 3, 4, 5.$$

Those satisfying the condition  $\Re(z)\Im(z) < 0$  (points in the second or fourth quadrant) are:

$$z_2 = -\sqrt{1/2} + i\sqrt{1/2}, \quad z_5 = \sqrt{1/2} - i\sqrt{1/2}.$$

#### Exercise 14

Solve the following equation in the complex variable  $z$ .

$$z^3 + 8 = 0$$

The equation to be solved is, clearly,

$$z^3 = -8$$

Setting  $z = r(\cos \vartheta + i \sin \vartheta)$  and taking into account that  $|-8| = 8$ ,  $\arg(-8) = \pi + 2k\pi$ ,  $k \in \mathbb{Z}$ , we obtain

$$\begin{cases} r^3 = 8 \\ 3\vartheta = \pi + 2k\pi \end{cases} \quad \begin{cases} r = 2 \\ \vartheta = (2k+1)\pi/3. \end{cases}$$

We therefore have the three roots

$$z_k = 2 \cos((2k+1)\pi/3) + 2i \sin((2k+1)\pi/3), \quad k = 0, 1, 2,$$

i.e.,

$$z_0 = 1 + \sqrt{3}i, \quad z_1 = -2, \quad z_2 = 1 - \sqrt{3}i.$$

#### Exercise 15

Solve the following equation in the complex variable  $z$ .

$$z^3 = -27i$$

Setting  $z = r(\cos \vartheta + i \sin \vartheta)$  and taking into account that  $|-27i| = 27$ ,  $\arg(-27i) = -\pi/2 + 2k\pi$ ,  $k \in \mathbb{Z}$ , we obtain

$$\begin{cases} r^3 = 27 \\ 3\vartheta = -\pi/2 + 2k\pi \end{cases} \quad \begin{cases} r = 3 \\ \vartheta = (4k-1)\pi/6. \end{cases}$$

We therefore have the three roots

$$z_k = 3 \cos((4k-1)\pi/6) + 3i \sin((4k-1)\pi/6), \quad k = 0, 1, 2,$$

i.e.,

$$z_0 = 3\sqrt{3}/2 - 3i/2, \quad z_1 = 3i, \quad z_2 = -3\sqrt{3}/2 - 3i/2.$$

### Exercise 16

Solve the following equation in the complex variable  $z$ .

$$z^5 + (3 + i\sqrt{3})z = 0$$

We have

$$z(z^4 + 3 + i\sqrt{3}) = 0$$

hence one solution is  $z = 0$ . The other four are found by solving

$$z^4 = -3 - i\sqrt{3}.$$

Setting  $z = r(\cos \vartheta + i \sin \vartheta)$  and taking into account that

$$|-3 - i\sqrt{3}| = 2\sqrt{3}, \quad \arg(-3 - i\sqrt{3}) = -5\pi/6 + 2k\pi, \quad k \in \mathbb{Z}$$

we obtain

$$\begin{cases} r^4 = 12^{1/2} \\ 4\vartheta = -5\pi/6 + 2k\pi \end{cases} \quad \begin{cases} r = 12^{1/8} \\ \vartheta = -5\pi/24 + k\pi/2. \end{cases}$$

Besides the solution  $z = 0$ , we therefore have four more roots

$$z_k = 12^{1/8} \cos((12k-5)\pi/24) + 12^{1/8} i \sin((12k-5)\pi/24), \quad k = 0, 1, 2, 3.$$

### Exercise 17

Solve the following equation in the complex variable  $z$ .

$$\left(\frac{z+i}{1+i}\right)^3 = -8i$$

Setting  $w = (z+i)/(1+i)$ , we solve  $w^3 = -8i$ . Writing  $w = r(\cos \vartheta + i \sin \vartheta)$  and taking into account that  $|-8i| = 8$ ,  $\arg(-8i) = -\pi/2 + 2k\pi$ ,  $k \in \mathbb{Z}$ , we obtain

$$\begin{cases} r^3 = 8 \\ 3\vartheta = -\pi/2 + 2k\pi \end{cases} \quad \begin{cases} r = 2 \\ \vartheta = (4k-1)\pi/6. \end{cases}$$

We therefore have the three roots

$$w_k = 2 \cos((4k-1)\pi/6) + 2i \sin((4k-1)\pi/6), \quad k = 0, 1, 2,$$

i.e.,

$$w_0 = \sqrt{3} - i, \quad w_1 = 2i, \quad w_2 = -\sqrt{3} - i.$$

Going back to the variable  $z$ , from  $z = (1+i)w - i$ , the three solutions of the given equation are

$$z_0 = \sqrt{3} + 1 + i\sqrt{3} - 2i, \quad z_1 = -2 + i, \quad z_2 = 1 - \sqrt{3} - i\sqrt{3} - 2i.$$

### Exercise 18

Solve the following equation in the complex variable  $z$ .

$$\left(\frac{z-i}{i}\right)^4 = -16$$

From  $i^4 = 1$  we have  $(z - i)^4 = -16$ . Setting  $w = z - i$ , we solve  $w^4 = -16$ . Writing  $w = r(\cos \vartheta + i \sin \vartheta)$  and taking into account that  $|-16| = 16$ ,  $\arg(-16) = -\pi + 2k\pi$ ,  $k \in \mathbb{Z}$ , we obtain

$$\begin{cases} r^4 = 16 \\ 4\vartheta = -\pi + 2k\pi \end{cases} \quad \begin{cases} r = 2 \\ \vartheta = -\pi/4 + k\pi/2. \end{cases}$$

We therefore have the four roots

$$w_k = 2 \cos((2k-1)\pi/4) + 2i \sin((2k-1)\pi/4), \quad k = 0, 1, 2, 3,$$

i.e.,

$$w_0 = \sqrt{2} - i\sqrt{2}, \quad w_1 = \sqrt{2} + i\sqrt{2}, \quad w_2 = -\sqrt{2} + i\sqrt{2}, \quad w_3 = -\sqrt{2} - i\sqrt{2}.$$

Going back to the variable  $z$ , from  $z = w + i$ , the four solutions of the given equation are

$$z_0 = \sqrt{2} - i\sqrt{2} + i, \quad z_1 = \sqrt{2} + i\sqrt{2} + i, \quad z_2 = -\sqrt{2} + i\sqrt{2} + i, \quad z_3 = -\sqrt{2} - i\sqrt{2} + i.$$

### Exercise 19

Solve the following equation in the complex variable  $z$ .

$$z^5 = -\bar{z}$$

The equation can be seen as

$$z^5 = (-1)\bar{z}$$

Setting  $z = r(\cos \vartheta + i \sin \vartheta)$  and taking into account that  $|\bar{z}| = r$ ,  $\arg(\bar{z}) = -\vartheta + 2k\pi$ ,  $k \in \mathbb{Z}$ , and recalling that  $-1 = e^{i\pi}$ , we obtain

$$\begin{cases} r^5 = r \\ 5\vartheta = \pi - \vartheta + 2k\pi \end{cases} \quad \begin{cases} r = 0, r = 1 \\ \vartheta = (2k+1)\pi/6. \end{cases}$$

At this point,

- for  $r = 0$ , we obtain  $z_0 = 0$ ;
- for  $r = 1$ , we obtain the roots  $z = \cos((2k+1)\pi/6) + i \sin((2k+1)\pi/6)$ ,  $k = 0, 1, 2, 3, 4, 5$ .

### Exercise 20

The equation

$$z^4 - 2z^3 + 4z^2 - 2z + 3 = 0$$

has  $z_1 = i$  as a solution. Denoting by  $z_2, z_3, z_4$  the other solutions, we have:

- (a)  $\Im(z_1 z_2 z_3 z_4) = 0$     (b)  $\Re(z_1 + z_2 + z_3 + z_4) = 0$   
(c)  $\Re((z_1 + z_2 + z_3 + z_4)^{-1}) = 1/2$     (d) none of the other answers is correct

The equation has real coefficients, hence the solutions come in conjugate pairs; in particular, another solution is  $z_2 = -i$ . It follows that the polynomial

$$p(z) = z^4 - 2z^3 + 4z^2 - 2z + 3$$

is divisible by  $(z - i)(z + i)$ , i.e., by  $z^2 + 1$ . Performing the division we obtain

$$p(z) = (z^2 + 1)(z^2 - 2z + 3)$$

hence among the solutions we also have the roots of

$$z^2 - 2z + 3 = 0$$

which are  $z_3 = 1 - i\sqrt{2}$ ,  $z_4 = 1 + i\sqrt{2}$ .

Summarizing, the solutions of the given equation are

$$z_1 = i, z_2 = -i, z_3 = 1 - i\sqrt{2}, z_4 = 1 + i\sqrt{2}.$$

Since  $z_1 + z_2 + z_3 + z_4 = 2$  and  $z_1 z_2 z_3 z_4 = 3$ , the correct answers are (a) and (c).

### Exercise 21

Let  $z_h$ ,  $h = 1, \dots, 6$  be the solutions of

$$z^6 - (8 - i)z^3 - 8i = 0.$$

We have:

- (a)  $\sum_{h=1}^6 \Re(z_h) = -1$     (b)  $\sum_{h=1}^6 \Im(z_h) = -1$   
(c)  $\sum_{h=1}^6 \Re(z_h) = 1$     (d)  $\sum_{h=1}^6 \Im(z_h) = 0$   
(e)  $\sum_{h=1}^6 \Re(z_h) = 2$     (f) none of the other answers is correct

It is convenient to factor it as

$$(z^3 - 8)(z^3 + i) = 0$$

and then solve the two equations

$$z^3 = 8, z^3 = -i$$

with the usual method illustrated several times in the solutions of the previous exercises. We obtain the values:

$$z_1 = 2, z_2 = -1 + i\sqrt{3}, z_3 = -1 - i\sqrt{3},$$

$$z_4 = \sqrt{3}/2 - i/2, z_5 = i, z_6 = -\sqrt{3}/2 - i/2.$$

The correct answer is (d).